

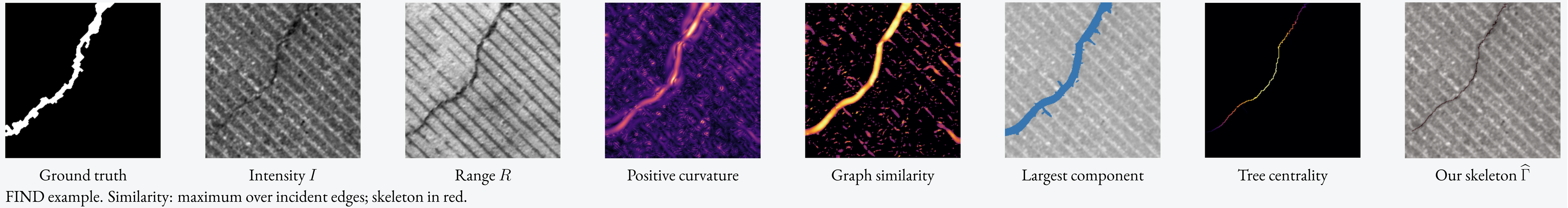
Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

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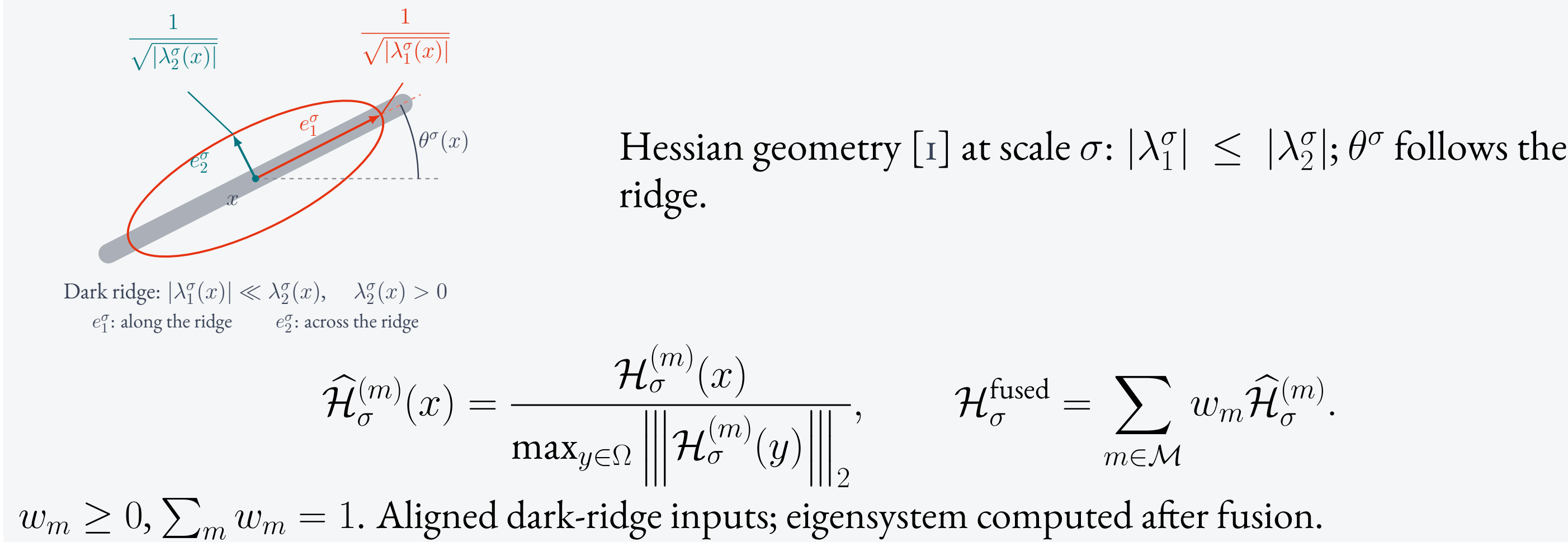
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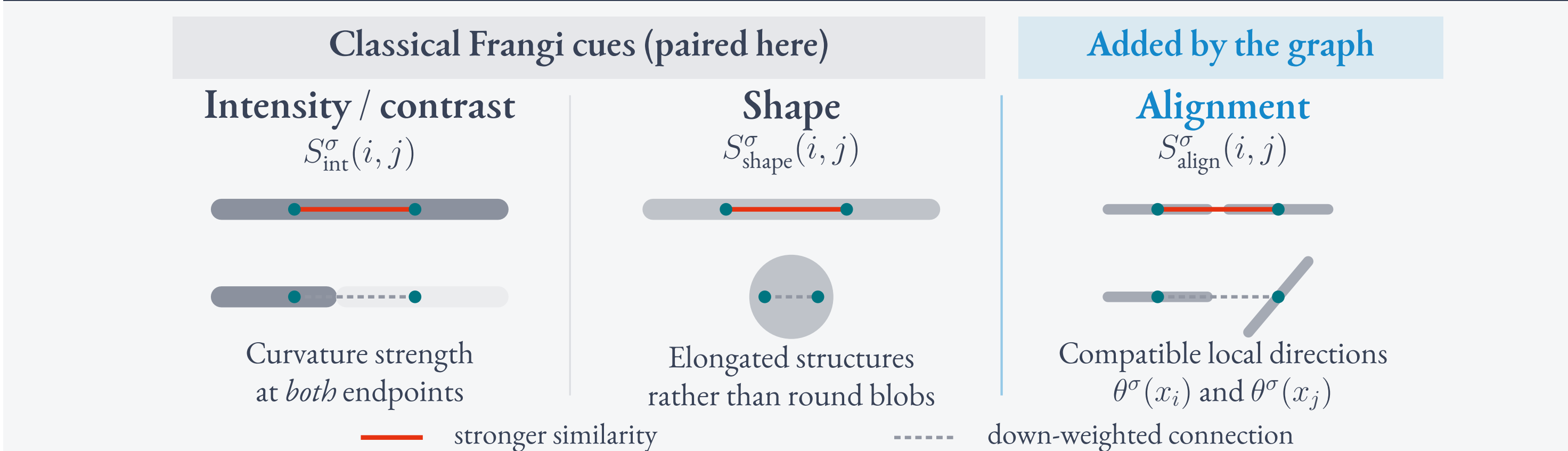
From multimodal images to a crack skeleton



1. Fuse the modalities



2. From pixel-wise Frangi to pairwise links



Classical Frangi cues + neighbor alignment. For pixels with $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

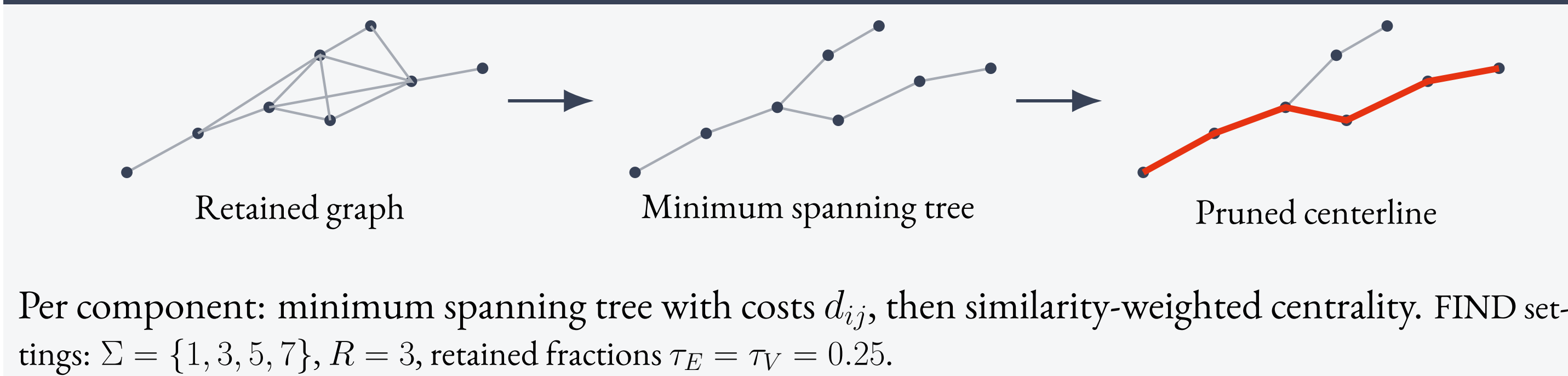
$$C_\sigma(x) = \max\{\lambda_2^\sigma(x), 0\}, \quad R_B^\sigma(x) = \frac{|\lambda_1^\sigma(x)|}{|\lambda_2^\sigma(x)|}.$$

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^\sigma(i, j) S_{\text{shape}}^\sigma(i, j) S_{\text{align}}^\sigma(i, j)$$

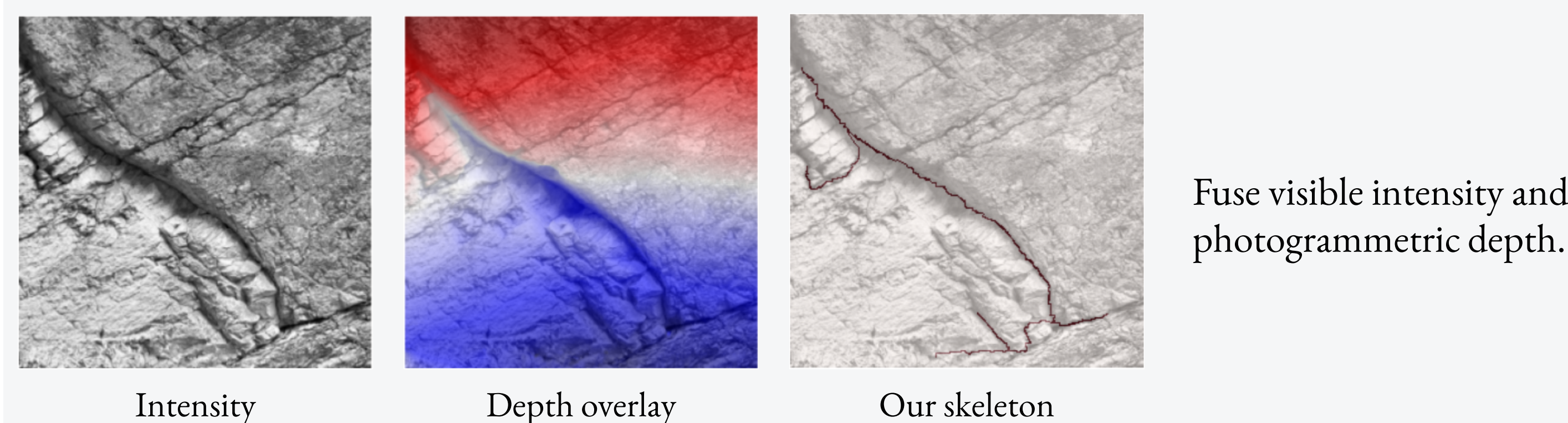
$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

ρ_{ij} penalizes long links; θ and $\theta + \pi$ denote the same axis.

3. Extract the centerline



Multimodality: Palais des Papes (France)



4. FIND: clean data

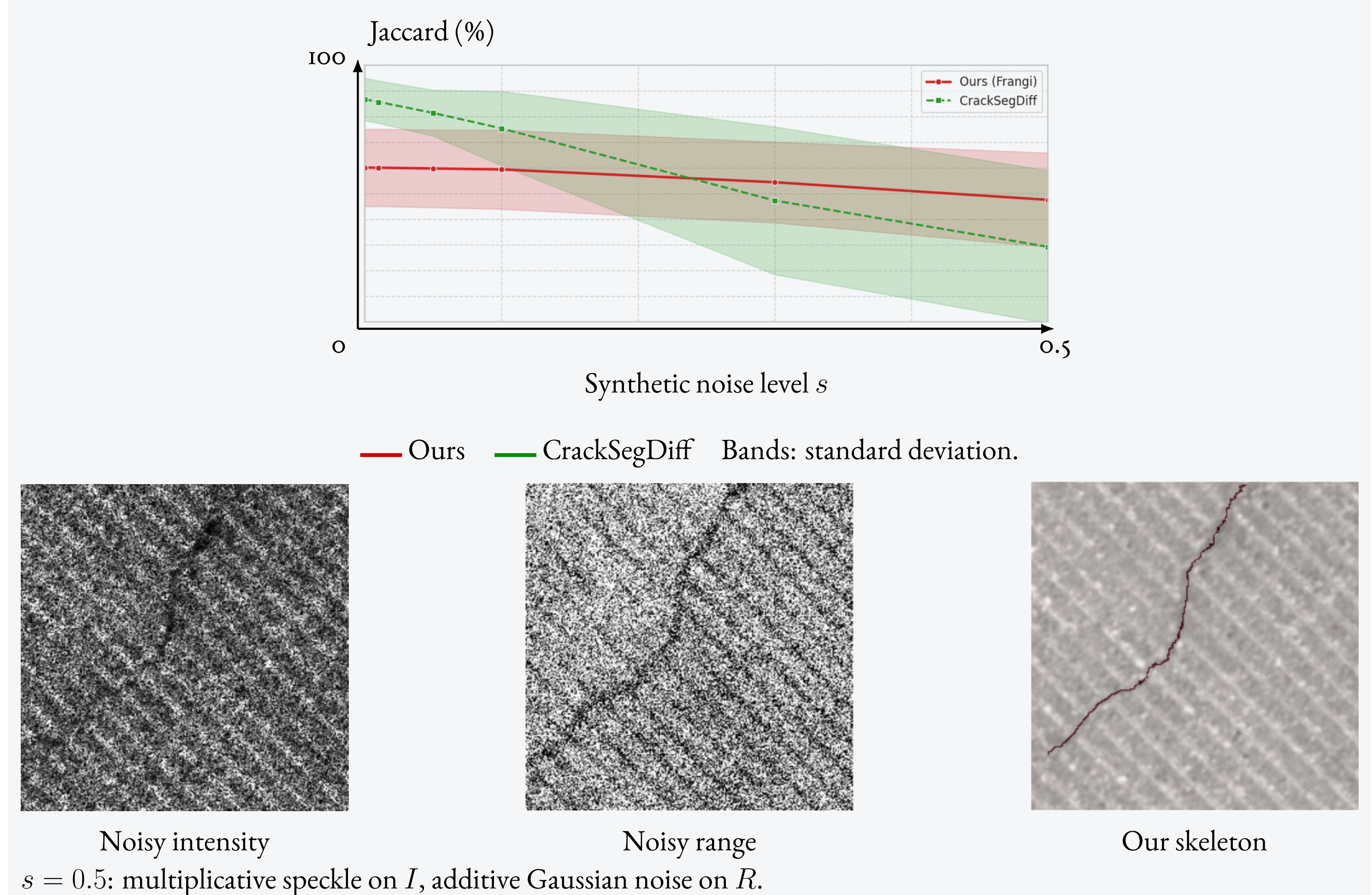
Method ($I + R$)	Jaccard $\uparrow$	Tversky $\uparrow$	$W \downarrow$
Ours, no training	60%	68%	18 px
CrackSegDiff [2]	87%	90%	5 px

FIND [3], 256×256 . Overlap: both skeletons dilated by 3 px; Tversky: $\alpha = 1, \beta = 0.5$; W : skeleton Wasserstein distance.

Fusion improves Jaccard: I : 41%; $I + R$: 60%; $I + R + F$: 63%. Equal weights; F is filtered range.

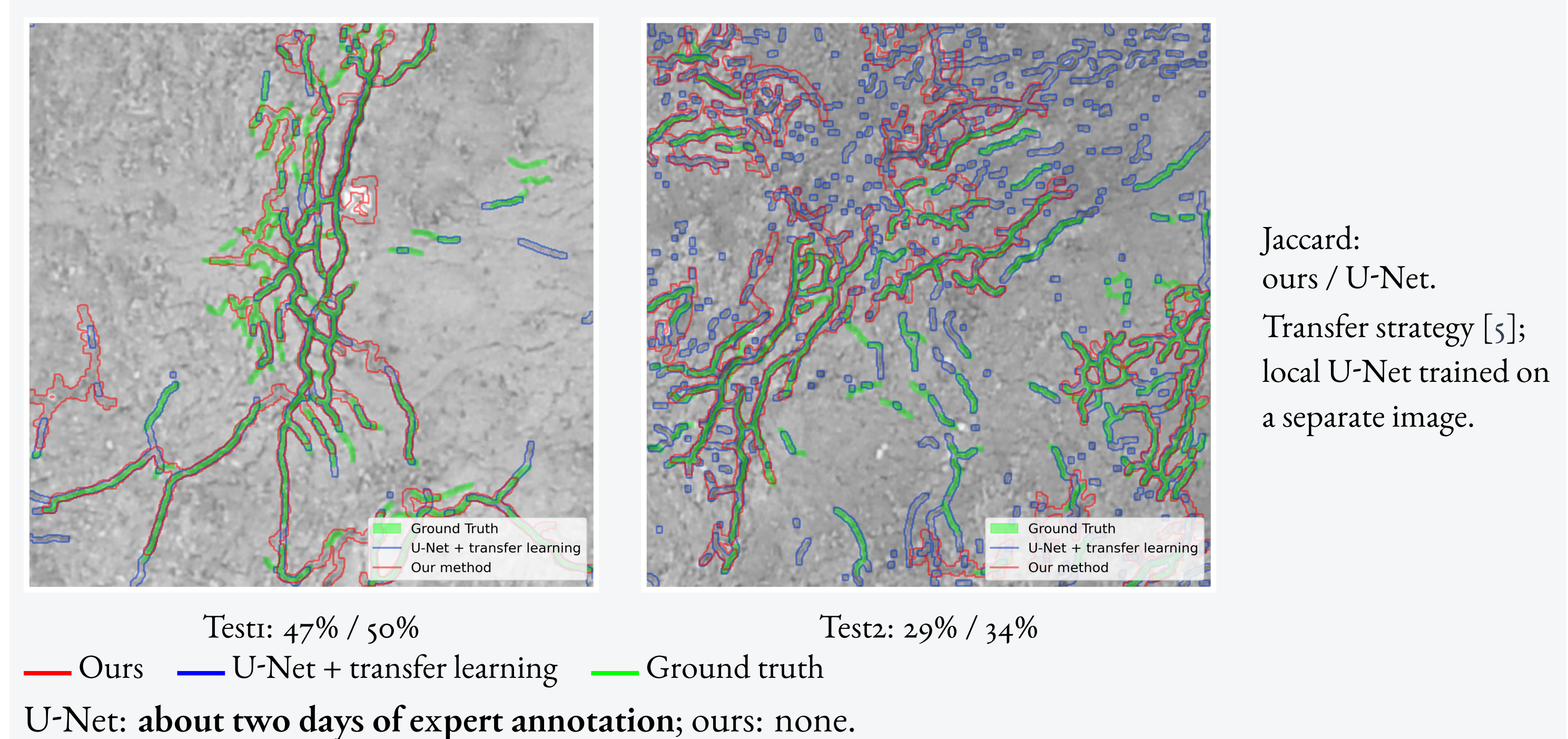
5. FIND: synthetic noise

Slower degradation under the tested synthetic noise.

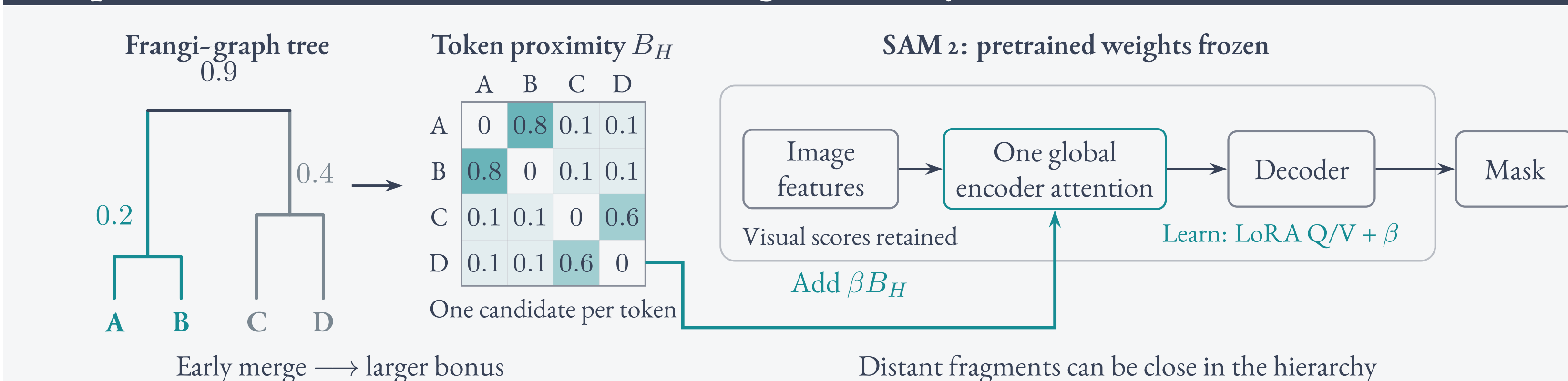


6. Geological domain shift [4]

Vaches Noires. Two Cerema UAV patches (512×512).



Perspective: frozen SAM 2 + LoRA + Frangi hierarchy



SAM 2 [6]: frozen pretrained weights.

LoRA [7] + one coefficient β : learned together.

$$A_H = \text{softmax}(L + \beta B_H).$$

L : visual scores; B_H : hierarchy bias.

Merge heights describe connectivity [8]. We use them to bias attention, following Graphormer [9]. CrackSAM [10] motivates adaptation to cracks.

Aim: fewer breaks without more false links.

References

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